

Hybrid Bayesian Optimization with Gradient-Based Local Refinement

Executive summary

The closest direct prior art to the workflow you described is not standard “gradient-based BO” in the usual sense of using gradients to optimize the **acquisition** function. Instead, the closest family consists of **BO-seeded local optimization** methods, where BO proposes a promising starting point for an expensive black-box objective and a local solver then refines that point on the **true objective**. The most relevant papers here are **BOWLS** by Gao, Yu, and Li, which uses BO to choose local-search start points and labels each start point with the **local optimum value** returned by the local search, and **BO-IPOPT** by Kyriakidis and coauthors, which uses EI to choose one or more BO candidates and then refines them with IPOPT, updating the surrogate using the refined local optima rather than the original EI candidates. A second, more indirect but still relevant line is **BLOSSOM**, which switches from BO to BFGS once the GP indicates a likely convex basin; and a third line is **local BO** methods such as **GIBO**, **probability-of-descent local BO**, **BayeSQP**, and **NeST-BO**, which learn local gradient or curvature information with a GP and then take gradient, Newton, or SQP-like steps. ¹

Across the primary sources reviewed, I did **not** find a canonical published method that exactly matches your proposed rule: “BO proposes x_{BO} , a gradient-based local optimizer refines it to x_{loc} , and if the two are sufficiently different, **both** points are added to the GP as separate observations.” BOWLS stores the **start point** but with the **local optimum value**, which is a different surrogate target; BO-IPOPT stores the **refined point(s)**, not both start and end points; BLOSSOM **switches** to local optimization rather than updating the GP with both points during hybrid steps; and local BO methods like GIBO / BayeSQP / NeST-BO internalize local optimization differently. That makes your dual-point update idea look less like a rediscovery of a standard recipe and more like a plausible, clearly motivated **variant or contribution**. ²

The main theoretical attraction of the dual-point update is straightforward: if x_{BO} and x_{loc} are materially different, adding both can reduce local posterior uncertainty more than adding only one point, and it can help the surrogate represent the geometry of the basin around the BO proposal rather than only the locally refined endpoint. The main risks are equally straightforward: the two points are usually **highly correlated**, they can cause GP conditioning problems if they are near-duplicates, they can over-concentrate the design in already promising basins, and with noisy observations they can produce overconfident posteriors unless duplicates/replicates and nuggets are handled carefully. The literature on EI convergence, local BO convergence, GP nuggets, noisy-input GPs, and heteroskedastic GP modeling all support that these issues are real and practically important. ³

My practical recommendation is to treat the dual-point update as a **conditional second-point acceptance** rule rather than as a default. In practice, keep both points only when at least one of the following is true: the kernel-scaled distance is large relative to learned GP lengthscales, the improvement $|y_{\text{loc}} - y_{\text{BO}}|$ is statistically meaningful under the observation noise model, or the conditional GP information gain from adding x_{loc} after x_{BO} is above a threshold. Otherwise, keep only the refined point, or treat the pair as a

replicate/near-replicate and aggregate them. This is especially important because classical BO theory is built around one expensive query per round, whereas local-search hybrids spend budget inside the local solver and therefore need more careful accounting to preserve exploration and sample efficiency. ⁴

Search strategy and scope

I prioritized primary and official sources in this order: original venue pages and accepted-paper PDFs from PMLR, NeurIPS, ICML, and official journal or author/lab pages; then arXiv and OpenReview versions where the official text was easier to access; then GitHub repositories from the authors or labs for code; and finally Google Patents for adjacent intellectual-property searches. Representative sources in the final synthesis include the original EGO and EI-convergence papers, the BO review literature, PMLR / NeurIPS papers for hybrid and local BO methods, author-hosted PDFs for BO-IPOPT and related control/process papers, and GitHub repos for GIBO, TuRBO, BayeSQP, NeST-BO, COBALT, BONNI, and BLOSSOM-related code. ⁵

A crucial scope distinction is that many BO papers and packages say they use gradient-based optimization, but they mean gradients of the **acquisition function**, not post-hoc local refinement of the expensive **objective**. The BO review literature explicitly treats acquisition optimization as an internal numerical subproblem that is usually much cheaper than querying the black-box objective. Your request is narrower: methods where BO chooses a candidate for the expensive problem and a local gradient-based solver then refines that candidate on the actual objective, potentially yielding additional GP training data. I therefore excluded papers that only use BFGS or L-BFGS to optimize EI, UCB, or other acquisitions, unless they also had a genuine BO-to-local-objective handoff. ⁶

Some details in your requested algorithm are unspecified in the literature and remain design choices: whether the true objective gradient is exact, noisy, automatic-differentiation based, adjoint-based, or estimated by finite differences; whether x_{BO} is explicitly evaluated before starting the local solver or is only the solver's initialization; whether gradients are budgeted equally with function evaluations; whether the domain is unconstrained or constrained; and whether the noise model is homoskedastic, heteroskedastic, or effectively noiseless. Where necessary below, I state assumptions explicitly rather than hiding them. ⁷

Prior art and taxonomy

A useful taxonomy starts with the **baseline BO line**. EGO introduced the modern GP-plus-EI template: fit a GP surrogate, maximize EI, evaluate the expensive objective, and update the GP. Later theoretical work showed that EI can converge under appropriate assumptions, but also cautioned that naive implementations can fail. This baseline matters because your proposal modifies exactly one part of that loop: what happens **after** the BO candidate is chosen and **before** the GP is updated. ⁸

The closest direct category is **BO as a global start-point selector for local search**. In **BOWLS**, BO chooses a start point x_n , a local search routine $L(f, x_n)$ returns (x_n^*, y_n^*) , and the GP is trained on (x_n, y_n^*) , meaning the start-point location is associated with the value of the local optimum reached from that start. In **BO-IPOPT**, the BO part uses EI to identify one or more candidates, which are then passed to IPOPT as starting points, and the dataset is updated using the resulting local optima. The BO-IPOPT papers are especially relevant because they explicitly discuss the fact that the refined local optima may differ significantly from the BO candidates, yet they still do **not** adopt your proposed “keep both points” rule. ⁹

A neighboring category is **switching or staged BO-to-local optimization**. **BLOSSOM** explicitly switches from BO to local BFGS once the GP indicates a likely convex region and sufficiently small global regret. This is conceptually close to your interest in refinement, but it is staged rather than per-iteration. Related work cited by GIBO also discusses switching from BO to local gradient methods once a good basin is identified.

10

A different but very important category is **local BO that learns derivatives or local structure and then takes local steps**. **GIBO** uses a GP over the objective to actively query points that shrink uncertainty in the gradient at the current iterate, then takes a gradient-based update. **Local BO via maximizing probability of descent** refines this idea by choosing samples that maximize the probability that the expected local step is a descent direction. Theoretical follow-up work by Wu, Kim, Garnett, and Gardner provides convergence guarantees for a prototypical local BO routine closely related to GIBO. **BayeSQP** and **NeST-BO** go further by using GP-inferred first- and second-order information to emulate SQP or Newton-like local optimization from zeroth-order data. These methods are not BO-seeded local refinement in the BOWLS / BO-IPOPT sense, but they are central to the modern landscape because they show how gradient-based local optimization can be tightly integrated with GP uncertainty.

11

A final adjacent group is **local or trust-region BO without explicit gradient refinement**. **TuRBO** uses multiple local trust regions with separate local models to scale BO to high dimensions and large budgets. The **Mixed-Global-Local kernel** paper similarly tries to encode local convex structure inside the surrogate itself. These methods are important as comparators because they already attack the same underlying problem—global exploration plus efficient local progress—but without the explicit two-stage “EI then objective-local-refinement” loop you asked about.

12

The most important take-away from this taxonomy is that the literature already contains several ways of combining BO and local methods, but the exact design choice you care about—**conditionally updating the GP with both the initial BO candidate and the locally refined point as distinct observations**—does not emerge as a standard default in the closest papers reviewed. That gap is the central novelty signal in your proposal.

1

Key papers and algorithm comparison

Core papers

Efficient Global Optimization of Expensive Black-Box Functions

Year: 1998. **Authors:** Donald R. Jones, Matthias Schonlau, William J. Welch. **Main idea:** introduce the EGO framework, using a GP surrogate and the Expected Improvement acquisition to trade off exploration and exploitation in expensive black-box optimization. **Why it matters here:** it is the canonical BO loop that your proposed hybrid modifies. **Algorithmic detail:** one GP update per expensive evaluation; no local refinement of the true objective. **Theory:** later EI-convergence results build directly on this framework.

13

Optimization, fast and slow: optimally switching between local and Bayesian optimization

Year: 2018. **Authors:** Mark McLeod, Stephen Roberts, Michael A. Osborne. **Main idea:** BLOSSOM switches between BO modes and a final local BFGS phase using GP-based convex-region and regret criteria. **Why it matters here:** it is a direct precedent for BO handing off to a gradient-based local solver on the actual objective. **Algorithmic detail:** BO uses predictive-entropy and global-regret-reduction acquisitions; once

the GP indicates a likely convex basin and small global regret, the method starts BFGS. It also uses the GP Hessian estimate to precondition the local phase. ¹⁴

Bayesian Optimization with Local Search

Year: 2020. **Authors:** Yuzhou Gao, Tengchao Yu, Jinglai Li. **Main idea:** BO chooses starting points for local search. **Algorithmic detail:** for each chosen start point x_n , a local search $L(f, x_n)$ returns a local optimum value y_n^* , and the GP is updated with (x_n, y_n^*) , not (x_n^*, y_n^*) . **Why it matters here:** it is the closest unconstrained hybrid ancestor of your proposal, but it uses a transformed target rather than separate start and refined observations. **Empirics:** BOWLS outperformed MLSL and EMS on most of the reported benchmark functions and on a logistic-regression example when cost was measured in combined function and gradient evaluations. ¹⁵

A hybrid algorithm based on Bayesian optimization and IPOPT and Enhanced hybrid algorithm based on Bayesian optimization and IPOPT for constrained optimization

Years: 2024 and 2025. **Authors:** Loukas Kyriakidis and coauthors. **Main idea:** BO provides globally promising start points; IPOPT performs local constrained refinement. **Algorithmic detail:** EI is evaluated to pick one or more BO candidates; IPOPT refines each candidate on the actual constrained problem; the updated GP receives the refined feasible local optima, not both the BO candidates and their refined endpoints. The enhanced version studies discrete-vs-continuous EI candidate generation, multiple candidates per outer iteration, and augmented-Lagrangian based constraint handling. **Why it matters here:** it is the closest constrained direct prior art and, importantly, it highlights the exact design choice your dual-point update would change. ¹⁶

Local policy search with Bayesian optimization

Year: 2021. **Authors:** Sarah Müller, Alexander von Rohr, Sebastian Trimpe. **Main idea:** GIBO actively queries points that reduce uncertainty in the gradient of the objective at the current iterate and then performs a gradient-based update. **Algorithmic detail:** the GP jointly models objective values and gradients; a custom acquisition minimizes one-step look-ahead gradient uncertainty; after those extra samples are collected, the method takes a gradient ascent or Adam step. **Why it matters here:** it is one of the cleanest examples of GP-guided local gradient optimization from zeroth-order observations. ¹⁷

The behavior and convergence of local Bayesian optimization

Year: 2023. **Authors:** Kaiwen Wu, Kyurae Kim, Roman Garnett, Jacob R. Gardner. **Main idea:** provide the first convergence analysis for a representative local BO routine closely related to GIBO. **Algorithmic detail:** the acquisition minimizes posterior uncertainty in the gradient; the algorithm then follows the posterior mean gradient. **Theory:** with noiseless observations and batch size $d + 1$, they show an $O(d/n)$ rate for squared gradient norm; with noise, convergence is substantially slower and the method requires growing batch sizes or repeated sampling near the current iterate. This paper is essential for thinking rigorously about your proposed dual-point variant. ¹⁸

BayeSQP: Bayesian Optimization through Sequential Quadratic Programming

Year: 2026. **Authors:** Paul Brunzema, Sebastian Trimpe. **Main idea:** use GP-inferred function values, gradients, and Hessians from zero-order data to construct an uncertainty-aware SQP-like local subproblem, followed by constrained Thompson-sampling line search. **Why it matters here:** it is a modern example of deep local-optimization structure living inside a BO framework, especially for constrained problems. **Empirics:** the paper reports advantages on specific high-dimensional constrained settings. ¹⁹

NeST-BO: Fast Local Bayesian Optimization via Newton-Step Targeting of Gradient and Hessian Information

Year: 2025 preprint, AISTATS 2026 code release. **Authors:** Wei-Ting Tang, Akshay Kudva, Joel A. Paulson. **Main idea:** target the local Newton step directly by actively learning gradient and Hessian information with a GP surrogate. **Theory:** under assumptions, the method inherits inexact-Newton convergence, with global progress and quadratic local rates once steps are accurate enough. **Why it matters here:** it is a strong contemporary reference for Hessian-aware local GP optimization, though it is not an EI-seeded two-stage hybrid. ²⁰

Comparison table

Published benchmarking is heterogeneous: some papers count only expensive objective evaluations, others count objective-plus-gradient calls, others compare regret or CPU time, and constrained methods often evaluate feasibility differently. A pooled numerical chart would therefore be more misleading than informative. The table below keeps each paper in its native benchmark context. ²¹

Method name	Year	Authors	Uses EI or other acquisition	Applies local gradient refinement	Updates GP with multiple points	Theoretical guarantees	Empirical benchmarks	Code
EGO / EI baseline ¹³	1998	Jones, Schonlau, Welch	EI	No	Usually one point per BO round	EI convergence under later assumptions; rates in Bull 2011	Classical expensive black-box benchmarks	Bay
BLOSSOM ²²	2018	McLeod, Roberts, Osborne	PES + Global Regret Reduction	Yes, staged BFGS after GP-based switching	No	Regret-based switching and stopping rationale, but not a standard no-regret BO theorem	Selection of optimization problems; claims superior convergence and principled stopping	gpb

Method name	Year	Authors	Uses EI or other acquisition	Applies local gradient refinement	Updates GP with multiple points	Theoretical guarantees	Empirical benchmarks	Coc
BOWLS ²³	2020	Gao, Yu, Li	Generic BO acquisition α ; not fixed to EI in the text	Yes, conjugate-gradient local search	No in your sense; GP gets start points labeled by local-optimum values	No formal global BO-style guarantee reported	Price, Branin, Cosine-mixture, Trid, Hartmann, Ackley, logistic regression; generally beat MSLSL/EMS in reported cost metric	No identifier price
BO-IPOPT family ¹⁶	2024, 2025	Kyriakidis et al.	EI	Yes, IPOPT	Yes, multiple refined optima can be added per outer iteration; not both proposal and refined endpoint	No formal global guarantee reported	Constrained test and energy-system problems; report faster convergence / fewer function evaluations than BOWLS and MS-IPOPT, with improved scaling in the enhanced paper	No identifier price
GIBO ¹⁷	2021	Müller, von Rohr, Trimpe	Gradient-Information acquisition, not EI	Yes, gradient ascent / Adam after GP-guided sampling	Yes, batch points for gradient learning	Convergence supplied later by Wu et al. 2023	Synthetic objectives and RL benchmarks; improved sample efficiency and lower variance than random perturbation baselines	gibo

Method name	Year	Authors	Uses EI or other acquisition	Applies local gradient refinement	Updates GP with multiple points	Theoretical guarantees	Empirical benchmarks	Code
Local BO via maximizing probability of descent ²⁴	2022	Nguyen, Wu, Gardner, Garnett	Probability of Descent	Yes	Yes, local sampling around current iterate	Closely tied to later local-BO convergence analyses	Synthetic and local-optimization settings	No identifier
BayeSQP ²⁵	2026	Brunzema, Trimpe	Constrained Thompson-sampling line search; uncertainty-aware SQP subproblem	Yes, SQP-like local refinement	No, typically one new query per iteration	Uncertainty-aware SOCP subproblem; no classical BO global-regret theorem claimed	Specific high-dimensional constrained black-box benchmarks	bayesqp
NeST-BO ²⁰	2025, 2026	Tang, Kudva, Paulson	One-step look-ahead Newton-step error bound	Yes, Newton-like local refinement	Yes, batch step-targeting samples	Inherits inexact-Newton style convergence under assumptions	Synthetic and real-world tasks including very high dimension	NeST-BO
Proposed dual-point EI-local-refinement variant	—	This report's synthesis	EI by default, but other acquisitions possible	Yes	Yes, conditionally both x_{BO} and x_{loc}	No published theorem known; see discussion below	No published benchmarks yet	Strategic proposal GPY IPO

Dual-point GP update analysis

The GP component of BO starts from the usual posterior equations. For a dataset $D_t = \{(x_i, y_i)\}_{i=1}^n$, covariance matrix K_t , and observation noise variance σ_n^2 , the posterior mean and variance at a candidate x are

$$\mu_t(x) = k_t(x)^\top (K_t + \sigma_n^2 I)^{-1} y_t, \quad s_t^2(x) = k(x, x) - k_t(x)^\top (K_t + \sigma_n^2 I)^{-1} k_t(x).$$

For minimization, one common EI form is

$$\text{EI}_t(x) = (f_t^{\min} - \mu_t(x) - \xi) \Phi(z_t) + s_t(x) \phi(z_t), \quad z_t = \frac{f_t^{\min} - \mu_t(x) - \xi}{s_t(x)}.$$

These are the mathematical objects your hybrid changes only indirectly, by augmenting D_t with one or two evaluations per BO-guided outer iteration. ²⁶

From a **bias/variance** perspective, adding both $(x_{\text{BO}}, y_{\text{BO}})$ and $(x_{\text{loc}}, y_{\text{loc}})$ is attractive when the two points carry meaningfully different information. In finite samples, it can reduce posterior variance around the proposed basin more than adding only the refined point, and it can prevent the GP from “forgetting” the part of the basin that EI itself judged promising. In that sense, the two-point update can make the local geometry between proposal and endpoint more visible to the surrogate. This is especially relevant when the local solver routinely moves some distance before finding a better basin-bottom point. That said, this is an inference from standard GP posterior structure and the hybrid methods reviewed, not a theorem already proved in the literature. ²⁷

The countervailing effect is **correlation and overconfidence**. If x_{loc} lies within the same learned lengthscale neighborhood as x_{BO} , the second point may add very little new information while still reducing local posterior variance sharply. In an exact or nearly noiseless GP, very close points can also make K_t ill-conditioned. The standard remedy is a nugget/noise term, but the nugget changes the emulator from exact interpolation to smoothing, so the user must interpret uncertainty and extrapolation accordingly. Heteroskedastic GP models and replicate-aware GP implementations are useful when repeated or near-repeated sampling becomes common. ²⁸

The most rigorous way to think about “should I keep both?” is via **conditional information gain**. Once x_{BO} has been added, the incremental information from the refined point is

$$\Delta I(x_{\text{loc}} \mid D_t \cup \{x_{\text{BO}}\}) = \frac{1}{2} \log \left(1 + \frac{s_{t+\text{BO}}^2(x_{\text{loc}})}{\sigma_n^2} \right),$$

where $s_{t+\text{BO}}^2(\cdot)$ is the posterior variance after adding x_{BO} . If this quantity is tiny, the refined point is mostly redundant as GP training data even if it is a better optimizer iterate. This criterion is not standard in BOWLS or BO-IPOPT, but it is a principled acceptance rule suggested by the GP information structure. ²⁹

For **convergence**, one must be careful. Classical global BO theory is built around the fact that each expensive evaluation is chosen by an exploration-preserving policy such as GP-UCB or EI under suitable assumptions. Local BO theory, in turn, studies repeated local uncertainty reduction and local descent. Your dual-point rule mixes the two. My best synthesis is this: if the algorithm continues to commit at least one expensive query per outer round to the BO acquisition and only adds the refined point as an **extra** query, then asymptotically it should not be worse, in an information sense, than the one-point method. But in finite budgets, it can absolutely become **more exploitative**, because the extra point tends to land in already-promising basins, and the associated hyperparameter refits can shorten GP lengthscales and encourage further local concentration. That makes global exploration weaker unless the acquisition retains a nontrivial exploration term or a trust-region / restart mechanism. This is an inference consistent with EI theory, GP-UCB theory, and local-BO theory, rather than a published theorem for the exact dual-point algorithm. ³⁰

Noise complicates this further. Wu et al. show that local BO in the noisy setting converges more slowly, often needing larger effective batch sizes or repeated sampling as one approaches stationarity. If both x_{BO} and x_{loc} are noisy and close, the observed improvement may be mostly noise. In that setting, it is safer to treat near-duplicate pairs as **replicates**, average them, estimate local noise, and fit a heteroskedastic or replicate-aware GP rather than pretending they are two cleanly independent locations. If inputs themselves are uncertain—for example because the local solver ends with approximate projection or stochastic steps—noisy-input GP models become relevant too. ³¹

The main practical pitfalls therefore cluster into four groups. **Overfitting / hyperparameter collapse** occurs when many candidate-endpoint pairs accumulate in a few attractive basins and marginal-likelihood fitting drives the GP toward very short lengthscales. **Correlated observations** reduce the effective design size and can make two-point updates mostly cosmetic. **Noise mishandling** makes small candidate-to-endpoint improvements look more significant than they are. **Computation** becomes nontrivial because exact GP fitting scales cubically in the number of retained points, and a dual-point update doubles the outer-loop data rate even before counting the hidden cost of the local solver itself. These concerns are exactly why practical hybrids often either keep only refined endpoints, transform start-point labels, or move to local/trust-region formulations. ³²

Proposed algorithm

I will refer to the report’s synthesized variant as **EI-LR-2U** for compactness. That is a descriptive label for this report, not a name from the literature.

Assumptions

The practical version below assumes a continuous bounded domain $\mathcal{X} \subset \mathbb{R}^d$, a differentiable GP kernel, and access to a local optimizer that can use either exact gradients, adjoint/automatic-differentiation gradients, or reliable finite-difference gradients. If gradients are unavailable or too noisy, the same architecture can still be used with a local derivative-free solver, but then it leaves the specific scope of your requested “gradient-based local refinement.” BO theory itself is classically derivative-free and global, while the local step is an extra modeling choice. ³³

Suggested acceptance rules

A robust implementation should not always keep both points. I recommend a three-part acceptance test:

$$\delta_{\Lambda}(x_{\text{BO}}, x_{\text{loc}}) = \sqrt{(x_{\text{loc}} - x_{\text{BO}})^{\top} \Lambda^{-2} (x_{\text{loc}} - x_{\text{BO}})}$$

where Λ is the diagonal matrix of GP lengthscales.

Accept both points if

$$\delta_{\Lambda} > \tau_x \quad \text{and} \quad |y_{\text{loc}} - y_{\text{BO}}| > \tau_y$$

or, more principled,

$$\Delta I(x_{\text{loc}} \mid D_t \cup \{x_{\text{BO}}\}) > \tau_I.$$

A practical default is to use an **OR** between the distance criterion and the information-gain criterion, plus a duplicate guard:

- if $\delta_{\Lambda} < 0.1$, treat the points as near-duplicates;
- if $0.1 \leq \delta_{\Lambda} < 0.5$, keep both only when the observed improvement is meaningful relative to noise;
- if $\delta_{\Lambda} \geq 0.5$, keeping both is usually justified unless the local point is infeasible or numerically unstable.

These thresholds are engineering recommendations, not literature defaults.

Pseudocode

Algorithm EI-LR-2U

Inputs:

- domain X
- expensive objective f
- local gradient-based optimizer LocalOpt
- GP prior $p(f)$ with kernel k and noise model
- acquisition $a_t(x) = \text{EI}_t(x)$ or another B0 acquisition
- thresholds τ_x, τ_y, τ_I
- initialization set D_0
- budget B

Initialize:

- $D \leftarrow D_0$
- fit GP on D

while budget remains:

- # B0 proposal
- $x_{B0} \leftarrow \arg\max_{\{x \in X\}} \text{EI}(x \mid D)$
-
- # Explicit evaluation at the B0 proposal if not already counted
- $y_{B0} \leftarrow f(x_{B0})$
-
- # Local gradient refinement on the true objective
- $x_{\text{loc}}, y_{\text{loc}}, \text{local_trace} \leftarrow \text{LocalOpt}(f, x_{B0})$
-
- # Optional feasibility / noise / duplicate handling
- if x_{loc} is infeasible:
- $x_{\text{loc}}, y_{\text{loc}} \leftarrow \text{repair_or_project}(x_{\text{loc}}, y_{\text{loc}})$
-
- # Compute acceptance diagnostics
- $d_{\text{scaled}} \leftarrow \text{kernel_scaled_distance}(x_{B0}, x_{\text{loc}}, \text{GP_lengthscales}(D))$
- $dy \leftarrow |y_{\text{loc}} - y_{B0}|$
- $\text{info_gain} \leftarrow \text{conditional_gp_information_gain}(x_{\text{loc}} \mid D \cup \{(x_{B0}, y_{B0})\})$
-
- if duplicate(x_{B0}, x_{loc}):
- $D \leftarrow D \cup \{\text{aggregate_or_replicate}((x_{B0}, y_{B0}), (x_{\text{loc}}, y_{\text{loc}}))\}$
- else if ($d_{\text{scaled}} > \tau_x$ and $dy > \tau_y$) or ($\text{info_gain} > \tau_I$):
- $D \leftarrow D \cup \{(x_{B0}, y_{B0}), (x_{\text{loc}}, y_{\text{loc}})\}$
- else:
- $D \leftarrow D \cup \{(x_{\text{loc}}, y_{\text{loc}})\}$ # or $\{(x_{B0}, y_{B0})\}$ depending on objective
- end if

```

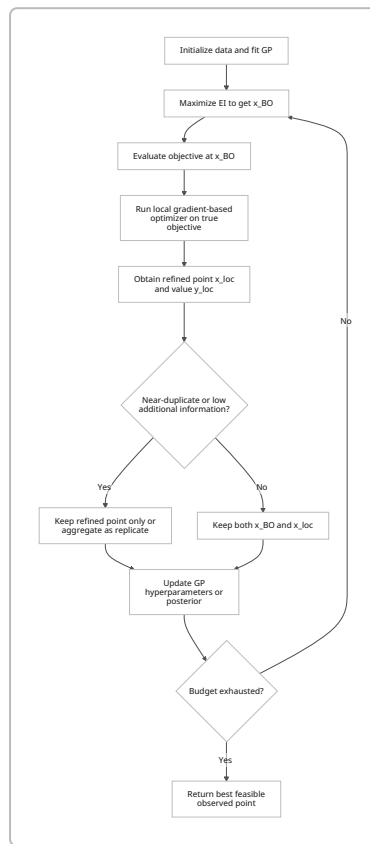
# Optional sparsified path update:
# add at most one or two intermediate local-trace points with maximal info
gain

    refit_or_update_GP(D)
end while

Return best observed feasible point in D

```

Mermaid flowchart



Practical implementation details and variants

If the local solver evaluates dozens of points internally, the cleanest compromise is **path sparsification**: do not dump the entire line-search or trust-region trace into the GP. Instead, keep the BO proposal, the final refined point, and optionally one intermediate point chosen by highest conditional information gain or largest predicted curvature change. This retains most of the geometric benefit without overwhelming the GP with highly correlated samples. This recommendation is especially compatible with the lessons from local BO and trust-region BO. ³⁴

For duplicates and near-duplicates, use one of three policies. In a near-noiseless simulator, **deduplicate** and keep the best numerically stable representative with a small nugget. In a noisy setting, **replicate-aware**

averaging is usually preferable. In a heteroskedastic setting, preserve replicates and use a model such as hetGP or another heteroskedastic GP. The wrong choice here is to keep many nearly identical points under a tiny fixed noise level, which tends to produce poor conditioning and overconfidence. ³⁵

For hyperparameters, avoid refitting the GP after every extra local point if your local refinement is aggressive. Instead, a stable practice is to update posterior state every iteration but refit kernel hyperparameters only every m outer rounds, or only when marginal likelihood changes materially. This is not a theorem; it is a practical hedge against the surrogate becoming too locally tuned too quickly. The BO review literature and the high-dimensional BO literature both emphasize how sensitive performance can be to GP hyperparameter behavior. ³⁶

Several sensible variants follow naturally. An **EI + trust-region-refinement** variant is useful when constraints or line-search-based solvers are awkward. A **qEI + parallel local refinement** variant is a natural extension of BO-IPOPT's multiple-candidate setting. A **gradient-enhanced dual-point GP** variant becomes attractive if objective gradients are already available, because then the GP can co-model values and derivatives rather than inferring all local structure from values alone. And a **cost-aware dual-point** variant can decide whether the extra refined endpoint is worth adding based on conditional information gain divided by incremental wall-clock or simulator cost. These variants line up well with the directions already explored by BO-IPOPT, GIBO, BayeSQP, and NeST-BO. ³⁷

Patents and open-source ecosystem

The patent landscape I found is mostly **adjacent**, not exact. A foundational patent, **US9864953B2**, covers systems and methods for BO with GP-based probabilistic models and standard acquisitions such as EI; it is relevant background IP, but it is not the specific BO-to-local-gradient-refinement workflow you described. Domain-specific patents include **US20240419863A1**, which uses BO with EI to select subsets of inputs that are then passed to a topology-optimization stage, and **WO2022058113A1**, which describes a GP/EI-style BO loop in a manufacturing optimization context. These documents show that BO, GP surrogates, EI, and downstream optimization stages are patent-active areas, but they did not surface an exact claim set matching the conditional dual-point GP update scheme. ³⁸

The open-source ecosystem is stronger for the **adjacent methods** than for the exact hybrid you proposed. Public repos are available for **GIBO**, **TuRBO**, **BayeSQP**, **NeST-BO**, **COBALT**, and **BONNI**, and a GitHub implementation exists for the code accompanying **Optimization, fast and slow**. These repositories collectively cover EI-style BO, local/trust-region BO, local gradient learning, SQP-like Bayesian local optimization, and interior-point hybridization, making them an excellent substrate for implementing and benchmarking your dual-point update variant. ³⁹

Concretely, the easiest engineering path is usually:

- **BoTorch / GPyTorch** or another GP-capable BO stack for the GP and EI layer,
- **SciPy**, **IPOPT**, or an SQP / trust-region solver for the local refinement layer,
- a small adapter that computes kernel-scaled distance, duplicate detection, conditional information gain, and replicate handling before the GP update.

The fact that BO-IPOPT and BONNI demonstrate the practicality of interior-point refinement, while GIBO, BayeSQP, and NeST-BO demonstrate the practicality of derivative-aware local surrogate logic, makes this a straightforward prototype even though it is not yet a standard library feature. ⁴⁰

Key references

Below are the references I would prioritize first if you are writing, implementing, or positioning this idea.

- Jones, Schonlau, and Welch, **Efficient Global Optimization of Expensive Black-Box Functions** (1998). Foundational EI paper. [Springer / DOI record](#). ⁴¹
- Bull, **Convergence Rates of Efficient Global Optimization Algorithms** (JMLR, 2011). Essential EI theory. [JMLR PDF](#). ⁴²
- Srinivas, Krause, Kakade, and Seeger, **Gaussian Process Optimization in the Bandit Setting** (ICML / arXiv 2009–2010). Core GP-UCB theory. [arXiv](#). ⁴³
- Frazier, **A Tutorial on Bayesian Optimization** (2018). Best high-level BO reference. [arXiv PDF](#). ⁴⁴
- Gao, Yu, and Li, **Bayesian Optimization with Local Search** (2020). Closest unconstrained direct ancestor. [arXiv PDF](#). ⁴⁵
- Kyriakidis et al., **A hybrid algorithm based on Bayesian optimization and IPOPT** (2024). Closest constrained direct ancestor. [DLR PDF](#). ⁴⁶
- Kyriakidis et al., **Enhanced hybrid algorithm based on Bayesian optimization and Interior Point Optimizer for constrained optimization** (2025). Best follow-up on candidate-selection and scaling choices. [DLR PDF](#). ⁴⁷
- McLeod, Roberts, and Osborne, **Optimization, fast and slow** (ICML 2018). BO-to-BFGS switching. [PMLR page](#). ⁴⁸
- Müller, von Rohr, and Trimpe, **Local policy search with Bayesian optimization** (NeurIPS 2021). GIBO. [NeurIPS PDF](#). ⁴⁹
- Wu, Kim, Garnett, and Gardner, **The Behavior and Convergence of Local Bayesian Optimization** (NeurIPS 2023). Best local-BO theory reference. [arXiv PDF](#). ⁵⁰
- Brunzema and Trimpe, **BayeSQP: Bayesian Optimization through Sequential Quadratic Programming** (2026). SQP-style GP local optimization. [PDF](#). ⁵¹
- Tang, Kudva, and Paulson, **NeST-BO** (AISTATS 2026 / preprint 2025). Newton-step-targeted local BO. [arXiv](#). ⁵²
- Rasmussen and Williams, **Gaussian Processes for Machine Learning** (2006). GP regression and derivative identities. [Official PDF](#). ⁵³
- Binois, Gramacy, and Ludkovski, **hetGP** (JSS, 2021). Important for replicate and heteroskedastic modeling. [JSS article](#). ⁵⁴
- McHutchon and Rasmussen, **Gaussian Process Training with Input Noise** (NeurIPS 2011). Important if local-refinement locations themselves are noisy/approximate. [NeurIPS PDF](#). ⁵⁵

¹ ² ⁷ ⁹ ¹⁵ ²¹ ²³ [arxiv.org](#)

<https://arxiv.org/pdf/1911.09159>

³ ³⁰ ⁴² [Convergence Rates of Efficient Global Optimization ...](#)

https://jmlr.org/papers/volume12/bull11a/bull11a.pdf?utm_source=chatgpt.com

⁴ ²⁹ ⁴³ [Gaussian Process Optimization in the Bandit Setting: No Regret and Experimental Design](#)

https://arxiv.org/abs/0912.3995?utm_source=chatgpt.com

5 8 13 Efficient Global Optimization of Expensive Black-Box ...

https://www.ressources-actuarielles.net/EXT/ISFA/1226.nsf/8d48b7680058e977c1256d65003ecbb5/f84f7ac703bf5862c12576d8002f5259/%24FILE/Jones98.pdf?utm_source=chatgpt.com

6 A Review of Bayesian Optimization

https://www.cs.ox.ac.uk/people/nando.defreitas/publications/BayesOptLoop.pdf?utm_source=chatgpt.com

10 14 22 Optimization, fast and slow: optimally switching between local and Bayesian optimization

https://arxiv.org/abs/1805.08610?utm_source=chatgpt.com

11 17 proceedings.neurips.cc

<https://proceedings.neurips.cc/paper/2021/file/ad0f7a25211abc3889cb0f420c85e671-Paper.pdf>

12 34 Scalable Global Optimization via Local Bayesian Optimization

<https://papers.neurips.cc/paper/8788-scalable-global-optimization-via-local-bayesian-optimization.pdf>

16 32 37 40 Enhanced hybrid algorithm based on Bayesian optimization and Interior Point OPTimizer for constrained optimization

https://elib.dlr.de/217039/1/Enhanced_BO-IPOPT.pdf

18 31 arxiv.org

<https://arxiv.org/pdf/2305.15572>

19 25 BayeSQP: Bayesian Optimization through Sequential Quadratic Programming

<https://publications.rwth-aachen.de/record/1029582/files/1029582.pdf>

20 52 NeST-BO: Fast Local Bayesian Optimization via Newton-Step Targeting of Gradient and Hessian Information

https://arxiv.org/abs/2510.05516?utm_source=chatgpt.com

24 Local Bayesian optimization via maximizing probability of ...

https://proceedings.neurips.cc/paper_files/paper/2022/file/555479a201da27c97aaeed842d16ca49-Paper-Conference.pdf?utm_source=chatgpt.com

26 27 53 Gaussian Processes for Machine Learning

https://gaussianprocess.org/gpml/chapters/RW.pdf?utm_source=chatgpt.com

28 The role of the nugget term in the Gaussian process method

https://arxiv.org/pdf/1005.4385?utm_source=chatgpt.com

33 44 A tutorial on bayesian optimization

https://arxiv.org/pdf/1807.02811?utm_source=chatgpt.com

35 The effect of the nugget on Gaussian process emulators of ...

https://www.asc.ohio-state.edu/statistics/comp_exp/jour.club/AndrianakisChallenor_CSDA2012.pdf?utm_source=chatgpt.com

36 Practical Bayesian Optimization of Machine Learning Algorithms

https://arxiv.org/abs/1206.2944?utm_source=chatgpt.com

38 Systems and methods for Bayesian optimization using ...

https://patents.google.com/patent/US9864953B2/en?utm_source=chatgpt.com

39 sarmueller/gibo: This repository contains the code for ...

https://github.com/sarmueller/gibo?utm_source=chatgpt.com

41 Efficient Global Optimization of Expensive Black-Box Functions

https://link.springer.com/article/10.1023/A%3A1008306431147?utm_source=chatgpt.com

45 arXiv:1911.09159v3 [stat.ML] 30 Jun 2020

https://arxiv.org/pdf/1911.09159?utm_source=chatgpt.com

46 A hybrid algorithm based on Bayesian optimization and ...

https://elib.dlr.de/207751/1/A%20hybrid%20algorithm%20based%20on%20Bayesian%20optimization%20and%20IPOPT.pdf?utm_source=chatgpt.com

47 Enhanced hybrid algorithm based on Bayesian ...

https://elib.dlr.de/217039/1/Enhanced_BO-IPOPT.pdf?utm_source=chatgpt.com

48 optimally switching between local and Bayesian optimization

https://proceedings.mlr.press/v80/mcleod18a.html?utm_source=chatgpt.com

49 Local policy search with Bayesian optimization

https://proceedings.neurips.cc/paper/2021/file/ad0f7a25211abc3889cb0f420c85e671-Paper.pdf?utm_source=chatgpt.com

50 The Behavior and Convergence of Local Bayesian ...

https://arxiv.org/pdf/2305.15572?utm_source=chatgpt.com

51 BayeSQP: Bayesian Optimization through Sequential Quadratic ...

https://publications.rwth-aachen.de/record/1029582/files/1029582.pdf?utm_source=chatgpt.com

54 hetGP: Heteroskedastic Gaussian Process Modeling and ...

https://www.jstatsoft.org/article/view/v098i13/1430?utm_source=chatgpt.com

55 Gaussian Process Training with Input Noise

https://proceedings.neurips.cc/paper/4295-gaussian-process-training-with-input-noise.pdf?utm_source=chatgpt.com